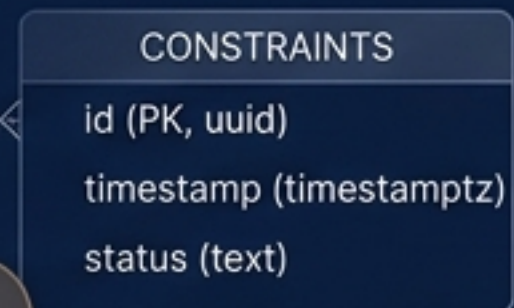
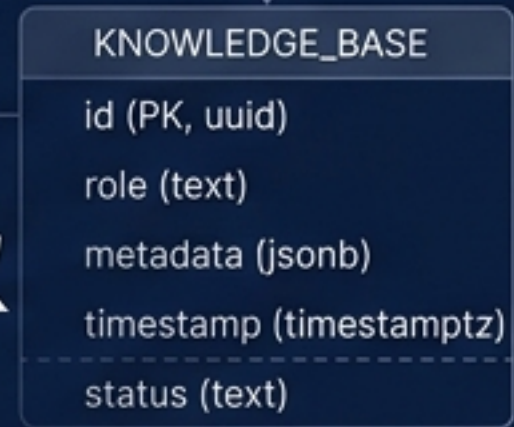
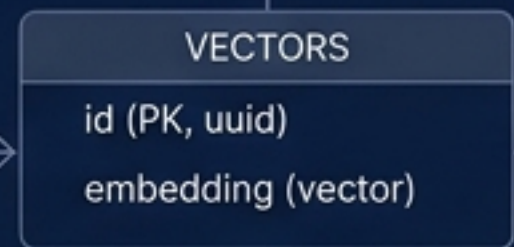


Clovbot Architecture & Systems Design

Engineered for zero-hallucination member assistance under strict regulatory and demographic constraints.



React 19

Node.js

Supabase (pgvector)

Azure OpenAI ⚠️

$$\text{Deflection} = S \times E \times R$$

Share (S)

Calls answerable without human-only context (~30-45%).

Engagement (E)

The highest elasticity lever.
46% of adults 50+ distrust AI.
30% of adults 65+ do not use the web.
Engagement is capped by UX.

Resolution (R)

Trust requires absolute citation accuracy and **zero hallucinations**.

Clovbot is architected specifically to maximize Engagement (Accessibility) and Resolution (Citation Accuracy) to achieve a realistic ~10% end-to-end deflection.



Typography & Touch

18px text floor and 44x44px touch targets. Exceeds WCAG AAA standards. Zero small, clustered targets.

Voice-First

Tap-to-start AND Press-and-hold (accommodating tremors/arthritis).

Native browser TTS fallback ensures the voice loop never runs out of credits.

Passwordless Auth

Pasteable 6-digit emailed OTPs directly inside the chat flow.

Prevents cognitive load and context loss (WCAG 3.3.8).

Bilingual Native

Spanish UI + Spanish retrieval from Spanish documents.

Never auto-translated, eliminating unverified claims.

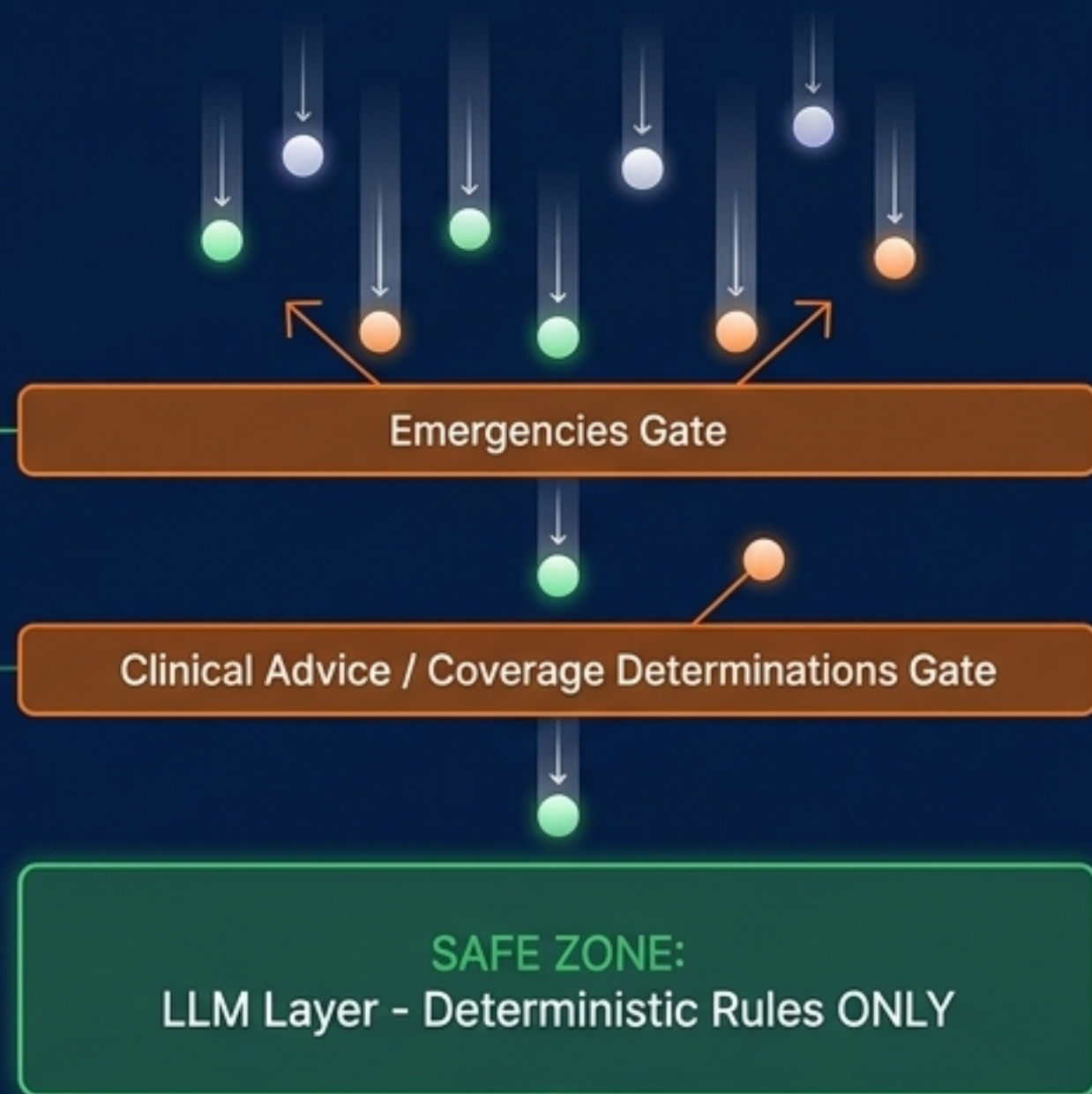
The Constraint

CMS-4201-F dictates AI must inform, not adjudicate.

The Solution

Implemented via deterministic rules, NOT LLM classifiers.

Zero-tolerance boundaries cannot rest on probability.



Emergencies Gate

Instantly breaks out to 911.

Clinical Advice / Coverage Determinations Gate

Refuses questions like "Was my denial correct?", names the exact boundary, and offers immediate pre-filled human handover. No silent routing failures.

Summary of Benefits

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Incorrect Layout Parsing (Text Merged Across Columns)

Summary of Benefits

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Precise 2-Column Extraction (Strict Gutter Separation)

Data Source

26 real 2026 Clover documents, fetched dynamically honoring robots.txt.

The Parsing Innovation

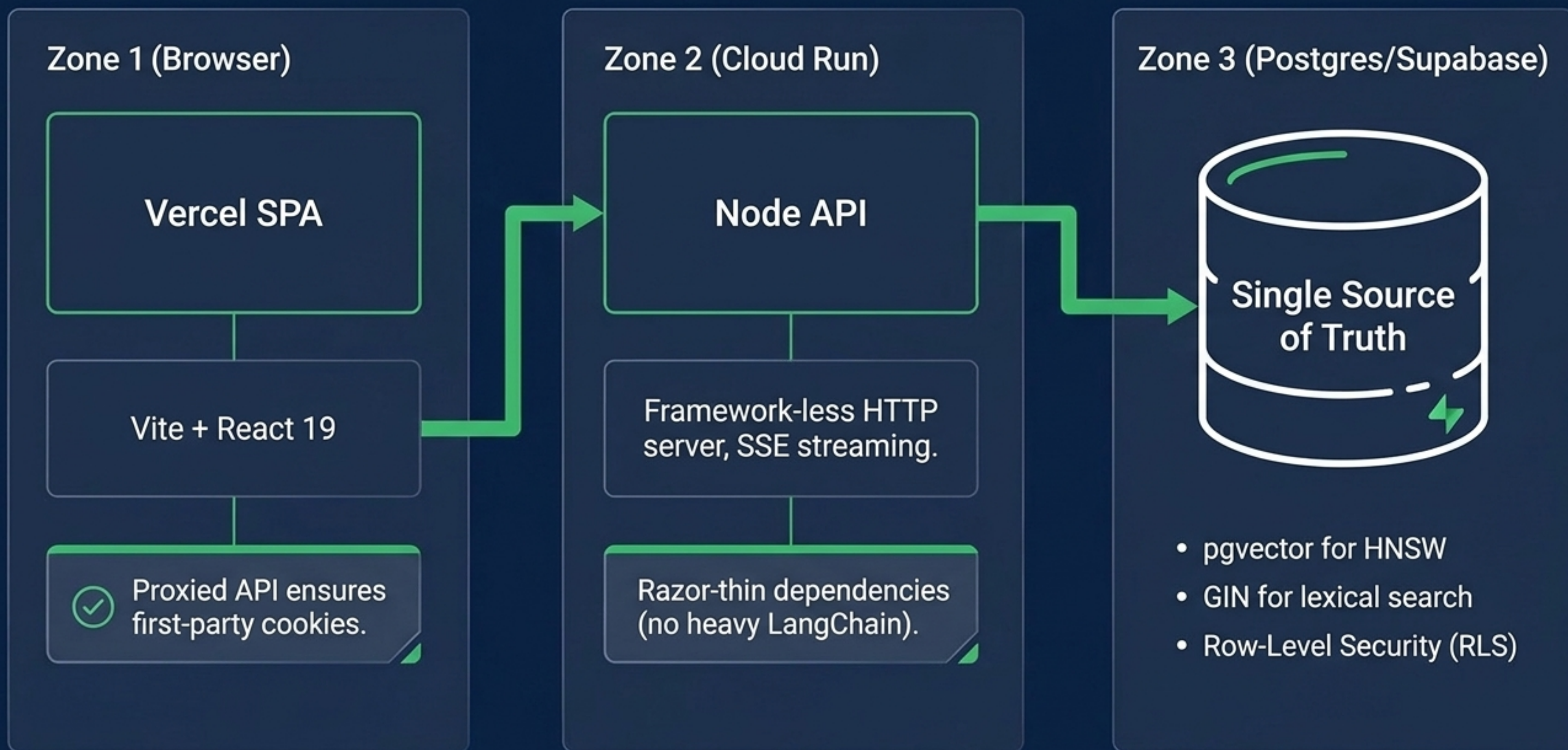
Parsed via Poppler using exact word coordinates. A page that cannot be attributed explicitly fails the build rather than guessing.

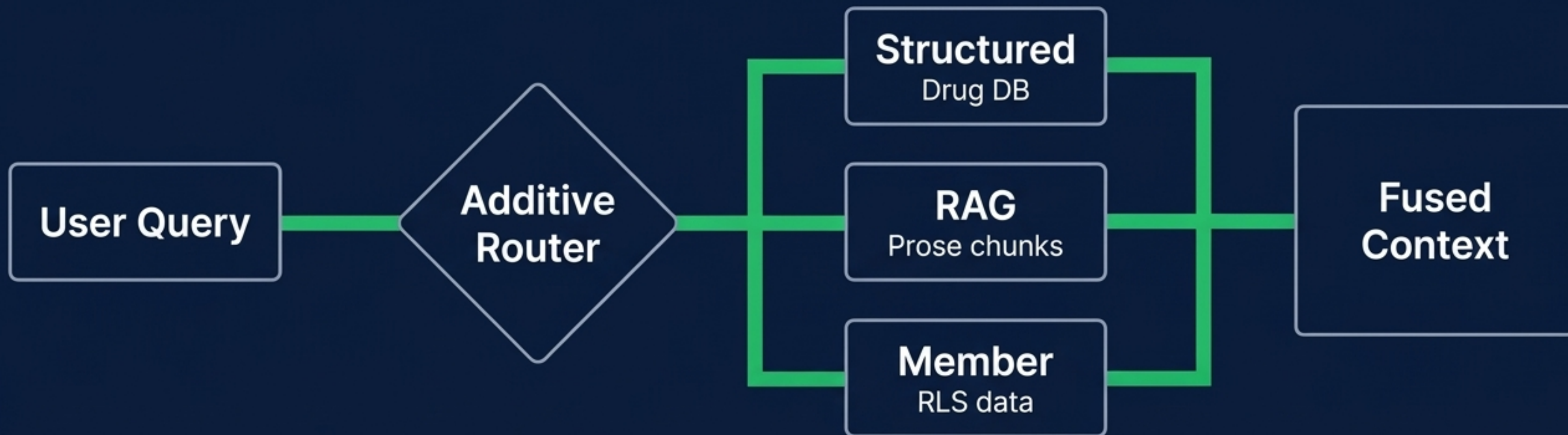
Contextual Chunking

Document, Plan, Year, and Section prefixes are appended to every chunk to drastically boost retrieval accuracy.

Typed Tables

The Formulary is parsed into 4,932 typed rows. Facts that live in tables get exact table queries, not fuzzy prose searches.





SQL-First Scoping

Plan ID and Language are filtered in SQL before ranking.

Prevents a highly-similar out-of-network plan from winning.

Reciprocal Rank Fusion (RRF)

Fuses Dense (HNSW) and Lexical (GIN) at k=60.

Lexical stems are strictly in the chunk's native language.

Additive Routing

A query like "Is Eliquis covered and how do I appeal?" routes to BOTH the typed drug table and prose RAG. Neither is dropped.

Standard RAG

Asks LLM for prose + citations.

Relies on prompt compliance.

Highly prone to **hallucinating** citations.

Clovbot Answer Contract

```
[
  {
    "claim": "Member's plan includes dental coverage with a $50 deductible.",
    "citationIds": ["plan_summary_doc_p4", "benefits_guide_2024_p12"]
  },
  {
    "claim": "Prior authorization is required for out-of-network specialists.",
    "citationIds": ["network_rules_v3_p7"]
  },
  {
    "claim": "The annual out-of-pocket maximum is $5,000.",
    "citationIds": ["plan_summary_doc_p2"]
  }
]
```

The App Validator

The Node API validates that every single claim is cited, and independently verifies that every cited ID was actually retrieved this turn.

The Result: Cite-or-refuse is a strict schema type, not an LLM instruction. An uncited claim fails schema validation and never reaches the screen.

1

Redact

Strip identifiers (SSN, ID) before the model.

2

Gate

Run deterministic CMS compliance rules.

3

Route

Additive selection across Drug tables, Docs, and User Data.

4

Retrieve & Rerank

RRF fusion + Local ONNX cross-encoder.
Absolute floor threshold applied (0.001).

5

Generate

Azure OpenAI strictly generates JSON facts.

6

Validate & Render

App maps citations, constructs prose, and streams to UI via SSE.

Database-Enforced Scoping

Trust boundary moved to Postgres.
Application uses a strict **NOBYPASSRLS** role.

The Failsafe

A query accidentally omitting a **WHERE member_id** clause returns exactly 0 rows, preventing cross-member data leakage.

API Node

Authenticated Request



Transaction-Local
Database Identity

Postgres Database Node

Data Store (RLS Enabled)

Append-Only Audit

Every authenticated read is
recorded in **member_access_log**
(Table.Column@RowID).

Data Integrity

V1 utilizes exclusively **synthetic member records**, guaranteeing **zero real PHI exposure**.

What is my specialist copay?

What is my out-of-network specialist copay?

**Semantically identical
but a \$10 difference**

The Rejection

Explicitly rejected
Semantic Caching.

The Reasoning

In healthcare benefits, a
single-word difference
entirely changes the
financial truth.

Loose similarity thresholds
return confidently wrong
amounts.

The Solution & Result

Postgres exact-keyed
caching based on
normalized question +
Plan ID + Snapshot.

Re-indexing clears cache
automatically for zero-cost
repeated answers.



Faithfulness



Structural



Routing



Login Detection

✓ The Golden Set

66 hand-labeled cases covering Answerable, Member-specific, Guardrails, Adversarial, and Spanish domains. 0.96 variance accounted for due to LLM helpful glossing.

⚠ Zero Tolerance Gates

0% False Negatives on Login detection (never guessing personal info without auth).
100% structural payload compliance. Misrouting a drug question instantly fails the build.

Lighthouse performance scorecard



Performance



Accessibility



Best Practices

Total Blocking Time: 0ms



Local vs. network

Zero third-party origins.
Proxied same-origin API.

ONNX reranker runs
server-side, never
loading in the browser.

On-Device Export

Conversations are rendered to PDF entirely client-side. PHI and claim amounts never go back over the wire for rendering.

**Member
Chatbot**

Agentic Handoff

V2 carries the entire conversation and context to the human agent, not just a summarized form.

MCP Server Integration

Exposes the exact RAG chunks the bot retrieved to the human agent's console to prevent re-deriving answers.

Bot-in-the-Room

Assistant stays in chat as a tool the human agent drives. The human makes the final determination, keeping CMS compliance intact.

**Agent-Assist
Framework**

Agentic Handoff

V2 carries the entire conversation and context to the human agent, not just a summarized form.

MCP-Tuning Loop

Exposes the exact RAG chunks the bot retrieved to the human agent's console to prevent re-deriving answers.

Fine-Tuning Loop

Future training relies on fully-cited payloads, not prose, utilizing the built-in ⚠️ rated-wrong feedback loop.

Pillar 1: Senior-First UX



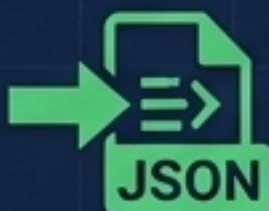
Built for the 65+ demographic constraint with voice-first accessibility and passwordless auth.

Pillar 2: Deterministic Guardrails



Hardcoded regulatory compliance that strictly informs and never adjudicates. ⚠️

Pillar 3: Zero-Hallucination Pipeline



Enforced by strict JSON schema typing, not LLM prompting.

Pillar 4: Enterprise RLS Database



Exact-keyed caching and Row-Level Security ensuring absolute data isolation.

Built to deflect calls by earning **member trust through absolute citation accuracy and strict regulatory compliance.**